

How long is a simulated double pendulum right?

Step size, floating-point precision, and the Lyapunov exponent

(Dated: September 19, 2026)

Abstract

Students who simulate a chaotic system learn that nearby initial conditions separate exponentially, but they seldom ask the corresponding question about the simulation itself, which is perturbed at every step by truncation and by roundoff. We describe a computational exercise that answers this question in about forty lines of code. The double pendulum is integrated with a fixed-step fourth-order Runge–Kutta method in arbitrary-precision arithmetic, so that the step h and the number of significand bits p can be varied independently, and we measure the time T for which each computed trajectory agrees with a far more accurate reference. The results follow a single formula in which the only property of the pendulum is the largest Lyapunov exponent λ . Until the other source of error takes over, halving h extends T by $4 \ln 2/\lambda$ and each additional bit extends it by $\ln 2/\lambda$, so the numerical error measures λ . The two values obtained this way, 0.35 ± 0.02 and 0.35 ± 0.01 in natural units, match the growth rate of a deliberately perturbed trajectory over the same interval and lie within 20% of the long-time exponent, 0.299. In double precision no choice of step keeps this pendulum accurate beyond about 101 natural time units, roughly 32 s for rods one meter long, and the computational work needed grows exponentially with T . Meanwhile, energy remains conserved to a part in 10^{10} long after the trajectory has become wrong.

I. INTRODUCTION

The double pendulum is the standard classroom example of chaos in a mechanical system, and this journal has treated it as one for thirty years. Shinbrot *et al.*[1] released two nearly identical physical pendulums together, watched them separate, and measured an exponential growth rate of $7.5 \pm 1.5 \text{ s}^{-1}$. Levien and Tan[2] built a double pendulum for the undergraduate laboratory and compared its motion with a numerical model. More recently, D’Alessio[3] combined analysis, simulation, and experiment, and computed the full spectrum of Lyapunov exponents. All three papers ask how the *physical* system amplifies an uncertainty in its initial state. Rafat *et al.*[4] and Calvão and Penna[5] studied the same system numerically.

A simulation amplifies errors in the same way, but its errors are introduced at every step rather than only in the initial condition. These errors are small and come from truncating the Taylor series of the solution and from rounding each arithmetic result to a finite number of bits, and each error is amplified from the step at which it is introduced. Hence, a student

who animates a double pendulum for a minute cannot tell whether the last forty seconds have anything to do with the solution of the equations that were typed in. The checks that students usually apply do not answer this question, since a trajectory that is entirely wrong can still conserve energy to many digits, as we show in Sec. IV E. Likewise, halving the step and seeing no visible change only tests the interval over which both runs are still accurate.

We describe an exercise that answers this question quantitatively, but none of the underlying facts is new. Lorenz[6] found sensitive dependence by re-entering rounded numbers into a running computation, and Lighthill[7] gave the horizon of predictability its name. The competition between truncation and roundoff is standard numerical analysis,[8] and the reliability of computed chaotic solutions has been studied in the research literature on atmospheric models[9] and with multiple-precision arithmetic.[10] In particular, Li *et al.*[11] showed that a given precision has a best step and a longest reliable time, and Kehlet and Logg[12] derived an error estimate of the form we use below. Wang and Li[13] found that the reliable time is linear in the number of digits, with a slope set by the Lyapunov exponent, and Mendes and Nepomuceno[14] estimated the exponent from the divergence of two equivalent computations that round differently. These studies treat the Lorenz and similar systems and are written for researchers. In this journal, Hayes[15] discussed what a computed chaotic trajectory means, and Faux and Godolphin[16] collected floating-point pitfalls for students. In an undergraduate thesis, Wild[17] integrated the double pendulum with RK4 at the IEEE significand widths and observed that the wider formats follow a 113-bit solution for longer, but did not relate this time to the Lyapunov exponent. However, we have not found a treatment at the level of an undergraduate course in which the student separates the two sources of error, measures each, and connects both to the Lyapunov exponent, so we give such a treatment here. The exercise teaches five things:

1. The time for which a simulation of a chaotic system is accurate depends only logarithmically on the error introduced per step, so large improvements in the integrator only add a small amount to that time.
2. Truncation error and roundoff error can be isolated from one another once the precision of the arithmetic is a parameter of the program rather than a property of the hardware.
3. Numerical error is itself a small perturbation, so its growth measures the Lyapunov exponent as well as a deliberately perturbed initial condition does.

4. For a given floating-point format there is a best step, and a longest accurate time that no step can improve upon.
5. Conservation of energy is not evidence that a chaotic trajectory is correct.

II. MODEL AND NUMERICAL METHOD

A. Equations of motion

Two point masses m hang in series from a fixed pivot on massless rigid rods of length l and move in a vertical plane under gravity. We measure the angles θ_1 and θ_2 from the downward vertical and time in units of $\sqrt{l/g}$, so the Lagrangian equations of motion contain no parameters:

$$\ddot{\theta}_1 = \frac{\omega_1^2 \sin \Delta \cos \Delta + \sin \theta_2 \cos \Delta + \omega_2^2 \sin \Delta - 2 \sin \theta_1}{2 - \cos^2 \Delta}, \quad (1)$$

$$\ddot{\theta}_2 = \frac{-\omega_2^2 \sin \Delta \cos \Delta + 2 (\sin \theta_1 \cos \Delta - \omega_1^2 \sin \Delta - \sin \theta_2)}{2 - \cos^2 \Delta}, \quad (2)$$

where $\omega_i = \dot{\theta}_i$ and $\Delta = \theta_2 - \theta_1$. The energy, in units of mgl , is

$$E = \omega_1^2 + \frac{1}{2}\omega_2^2 + \omega_1\omega_2 \cos \Delta - 2 \cos \theta_1 - \cos \theta_2. \quad (3)$$

Every run starts from rest with both rods horizontal, $\theta_1 = \theta_2 = \pi/2$, so $E = 0$. One natural time unit is 0.319 s when $l = 1$ m.

B. Integrator and arithmetic

We write Eqs. (1) and (2) as a first-order system $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y})$ for $\mathbf{y} = (\theta_1, \theta_2, \omega_1, \omega_2)$ and advance it with the classical fourth-order Runge–Kutta method (RK4) at a fixed step h . The exercise requires a fixed step, since an adaptive integrator adjusts h to hold its estimate of the local error near a tolerance, which hides the dependence we wish to expose.

All arithmetic uses the MPFR library,[18] which rounds every operation correctly to a significand of p bits, where the program chooses p . We call MPFR from Python through the `gmpy2` package, and Julia’s built-in `BigFloat` type wraps the same library and serves equally well. The values $p = 24, 53, 64$, and 113 reproduce the significand widths of IEEE single,

double, extended, and quadruple precision.[19] The 256-bit format has $p = 237$, which we treat as exact arithmetic, since its rounding unit is $2^{-237} \approx 10^{-71}$.

We choose steps $h = 2^{-k}$ because a power of two is represented exactly at every precision, so lowering p changes only the arithmetic. A decimal step such as 10^{-4} would be rounded differently at each precision, so runs intended to differ only in their arithmetic would also use slightly different steps. Wild[17] observed the effect of this choice, since replacing $h = 10^{-4}$ by $h = 2^{-13}$ kept the 53-bit and 64-bit solutions close to the 113-bit solution for longer.

C. Reference solution and measure of error

The error of a run is measured by the distance $d(t)$ between its lower bob and that of a reference run, in units of l . Since both bobs lie within $2l$ of the pivot, d saturates near 4 once two runs are unrelated. We define the *horizon* T of a run as the first time at which $d \geq 10^{-2}$, which is roughly when two superimposed animations become visibly different.

No exact solution is available, so the reference is a much more accurate run, and we show how far it can be trusted. Let e_a and e_r be the true errors of a run and the reference. By the triangle inequality, the measured separation satisfies $|d - e_a| \leq e_r$. All errors are amplified at the same rate, so the ratio e_r/e_a stays close to its initial value, which for two RK4 runs with steps 2^{-k_r} and 2^{-k_a} is $2^{-4(k_r - k_a)}$. Hence, against a reference with $k_r = 18$, the separation gives the true error of a run with $k_a \leq 16$ to better than one part in 250, as long as the run itself is accurate.

D. Three experiments

Truncation alone. With $p = 237$, we vary k from 4 to 17 and compare each run with the reference $k = 18$. Roundoff is about fifty orders of magnitude below the truncation error and plays no part.

Roundoff alone. With h fixed, we vary p from 11 to 128 and compare each run with the run at $p = 237$ and the same h . Both runs iterate the same discrete map, so the truncation error is common to them and cancels exactly, which leaves only the effect of rounding. We use a multiple-precision library because hardware arithmetic does not allow this separation.

Both together. With $p = 24, 32, 40, 53$, and 64, we vary k and compare with the reference

of the first experiment. This is the situation of a student working in ordinary single or double precision.

III. THE HORIZON OF PREDICTABILITY

Let $\mathbf{e}_n$ be the error after n steps. One step maps this error to $\mathbf{e}_{n+1} = \mathbf{J}_n \mathbf{e}_n + \boldsymbol{\tau}_n + \boldsymbol{\rho}_n$, where $\mathbf{J}_n$ is the Jacobian of the flow across the step. The step introduces the local truncation error $\boldsymbol{\tau}_n = O(h^5)$ of RK4 and the rounding error $\boldsymbol{\rho}_n = O(2^{-p})$. Unrolling the recursion expresses the error at time t as a sum over all earlier steps of the error introduced there, carried forward by the product of the intervening Jacobians. In a chaotic system, that product grows on average as $e^{\lambda(t-t_j)}$, where λ is the largest Lyapunov exponent. Hence, the sum is a geometric series dominated by its earliest terms, and

$$e(t) \approx (A h^4 + B 2^{-p} h^{-\gamma}) e^{\lambda t}. \quad (4)$$

The first term collects $1/h$ truncation errors of size h^5 per unit time. The second collects $1/h$ rounding errors of size 2^{-p} per unit time, which add like a random walk if their signs are independent, giving $\gamma = \frac{1}{2}$, and linearly otherwise, giving $\gamma = 1$. [20] Kehlet and Logg[12] derive $\gamma = \frac{1}{2}$. The constants A and B depend on the trajectory but not on h or p .

Setting $e(T) = \epsilon$ for a tolerance ϵ gives the horizon,

$$T = \frac{1}{\lambda} \ln \frac{\epsilon}{A h^4 + B 2^{-p} h^{-\gamma}}. \quad (5)$$

Everything in the exercise follows from Eq. (5). When truncation dominates and $h = 2^{-k}$,

$$T = T_A + \frac{4 \ln 2}{\lambda} k, \quad (6)$$

so each halving of the step extends the horizon by $4 \ln 2 / \lambda$, where the factor of 4 is the order of the method. When roundoff dominates,

$$T = T_B + \frac{\ln 2}{\lambda} (p - \gamma k), \quad (7)$$

so each additional bit extends the horizon by $\ln 2 / \lambda$, while each halving of the step now *shortens* the horizon. The two regimes meet where the two terms of Eq. (4) are equal, at $k^* = [p + \log_2(A/B)] / (4 + \gamma)$, and the horizon there,

$$T_{\max}(p) \approx \text{const} + \frac{\ln 2}{\lambda} \frac{4p}{4 + \gamma}, \quad (8)$$

is the longest attainable with p bits and a fourth-order method. Both slopes are properties of the pendulum alone, so we can measure λ from the numerical error.

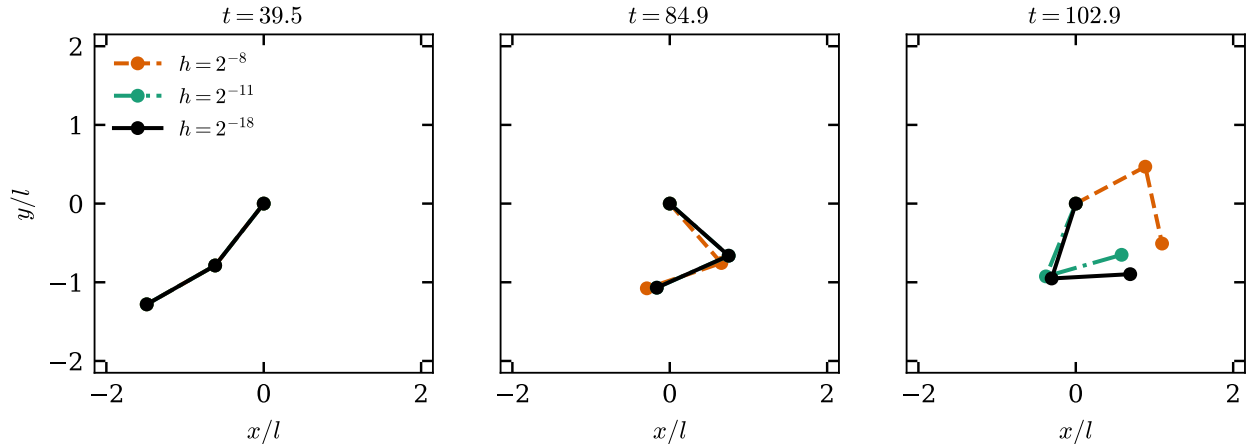


FIG. 1. Three double pendulums released from the same state and integrated with the same 237-bit arithmetic, differing only in the step h . The run with $h = 2^{-8}$ leaves the others near $t = 79$, and the run with $h = 2^{-11}$ near $t = 97$. Times are in units of $\sqrt{l/g}$.

IV. RESULTS

A. Step size

Figure 1 shows three pendulums integrated from identical states with identical 237-bit arithmetic, differing only in h . They coincide to within a line width for a long time and then separate in order of decreasing step. A supplementary animation, which draws the error curves alongside the evolving pendulums, shows this separation to a class faster than any graph.

Figure 2(a) shows $d(t)$ for a selection of steps. Each curve rises through many orders of magnitude at the same average rate and then saturates. Figure 2(b) shows why the curves are evenly spaced, since multiplied by h^{-4} they collapse onto a single curve, to within 0.21 decades for every $h \leq 2^{-8}$ wherever none has saturated, and the error at $t = 10$ scales as $h^{4.000}$. This curve is the amplification factor of the trajectory itself. To confirm this, we perturbed $\theta_1(0)$ by 2^{-170} , a perturbation small enough to remain in the linear regime throughout the run, and plotted the separation as the thin line in Fig. 2(b). This line follows the collapsed error curves through every plateau and burst. The integrator's error grows exactly as a perturbation of the initial condition does, since it is itself a perturbation.

The growth is exponential only on average, since the local rate fluctuates as the pendulum passes through more and less unstable parts of its motion, and these fluctuations are as much

a part of the lesson as the mean.

The horizons are plotted against k in Fig. 3(a). A straight line fitted for $7 \leq k \leq 16$ has slope 7.9 ± 0.4 time units per halving of the step, which Eq. (6) converts to $\lambda = 0.35 \pm 0.02$. The scatter about the line comes from the fluctuating local growth rate rather than noise. A horizon that falls just before a burst of growth gains little from the next halving, and one that falls just after gains a great deal. The gray staircase in Fig. 3(a) makes this quantitative, since it is the horizon predicted for every h from the measured growth of the 2^{-170} perturbation, where the single constant A of Eq. (4) is taken from the collapse. The staircase passes through the measured horizons step for step.

B. Precision

Figure 4 shows the second experiment for $h = 2^{-12}$. The picture is the same with 2^{-p} in place of h^4 (evenly spaced curves rising at a common rate). The horizons in Fig. 3(b) give a slope of 1.99 ± 0.06 time units per bit, which Eq. (7) converts to $\lambda = 0.35 \pm 0.01$. Changing the step by a factor of 256 moves the line only slightly. Comparing the errors at fixed early times for $h = 2^{-8}$ and 2^{-16} gives $\gamma \approx 0.6$ in Eq. (4), with values between 0.3 and 0.8 for individual precisions, which is close to the random-walk value.

We exclude the smallest precisions from the fit because with $p = 11$ and $h = 2^{-12}$ the pendulum *never moves*, and the way it fails is worth explaining to students. Wild[17] observed the same constant solution with eleven bits and $h = 10^{-4}$. The cause is not underflow, since the exponent range of MPFR is never approached. The velocity ω_1 at first falls by exactly h at every step for 2048 steps. At $t = \frac{1}{2}$ it reaches $-\frac{1}{2}$, where h is half a unit in the last place of ω_1 , so the sum is a tie, round-to-even returns $-\frac{1}{2}$, and the velocity is frozen. From then on the increment to the angle is $h|\omega_1| = 2^{-13}$, a quarter of the 2^{-11} needed to change the last of eleven bits in $\theta_1 = \pi/2$. Hence, the computed pendulum hangs horizontal forever with a constant velocity that produces no motion. With the same eleven bits and $h = 2^{-8}$ the velocity would not freeze until $|\omega_1| = 8$, the increment $h|\omega_1|$ first exceeds 2^{-11} at $t = 0.129$, and the pendulum falls. More generally, for this release the velocity freezes at 2^{p-k} and the angle needs $|\omega_1| > 2^{k-p}$, so the pendulum stalled for every $h < 2^{-p}$ that we tried and moved for every $h > 2^{-p}$. At $h = 2^{-p}$ the increment is itself a tie, so the outcome depends on the last bit of $\pi/2$. Here a smaller step has made the error

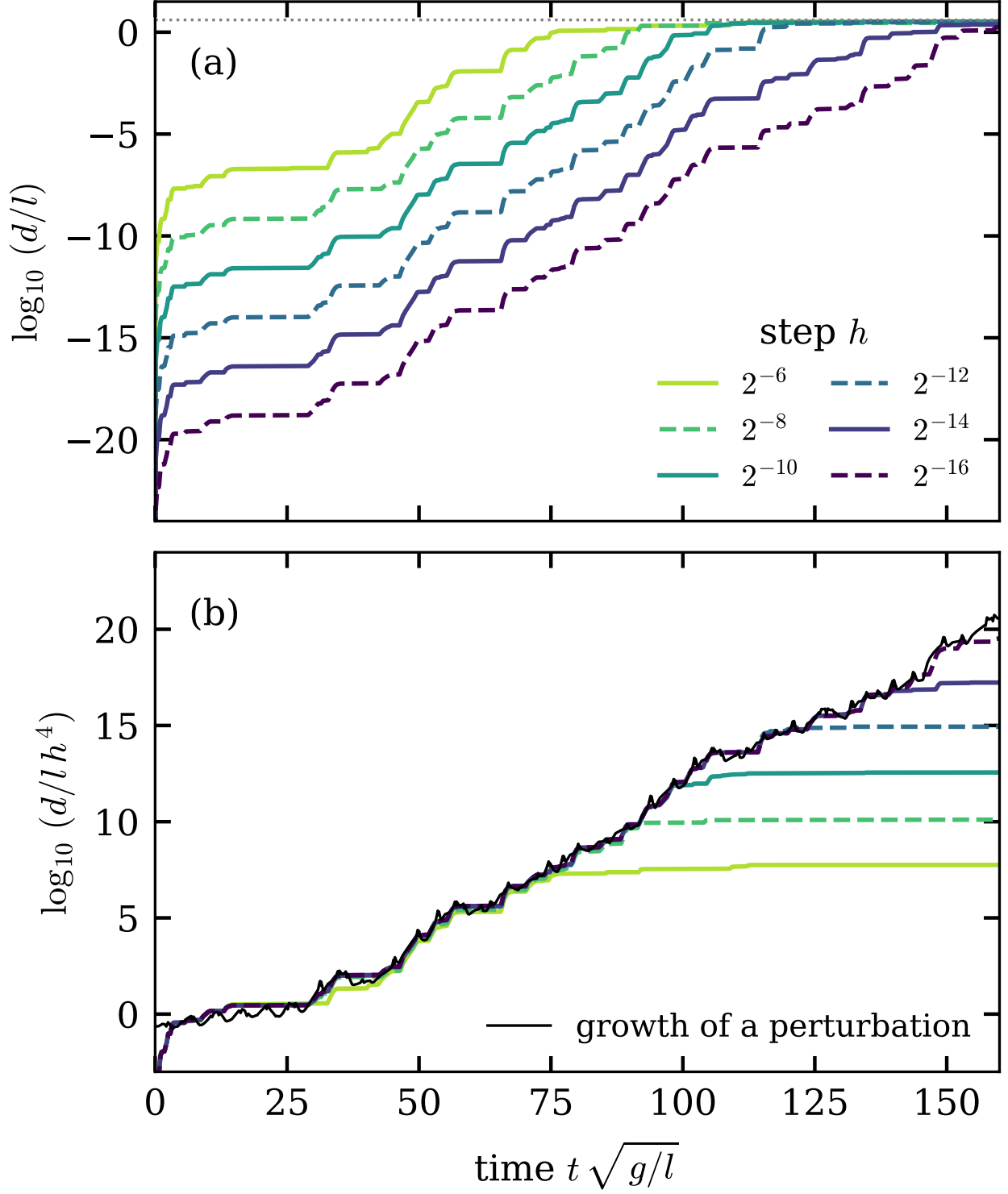


FIG. 2. (a) Running maximum of the separation d of the lower bob from that of the reference run ($h = 2^{-18}$) for six values of the step, in 237-bit arithmetic. The dotted line marks the largest possible separation, $4l$. (b) The same curves multiplied by h^{-4} collapse. The thin black line is the separation of two runs whose initial conditions differ by 2^{-170} , shifted vertically.

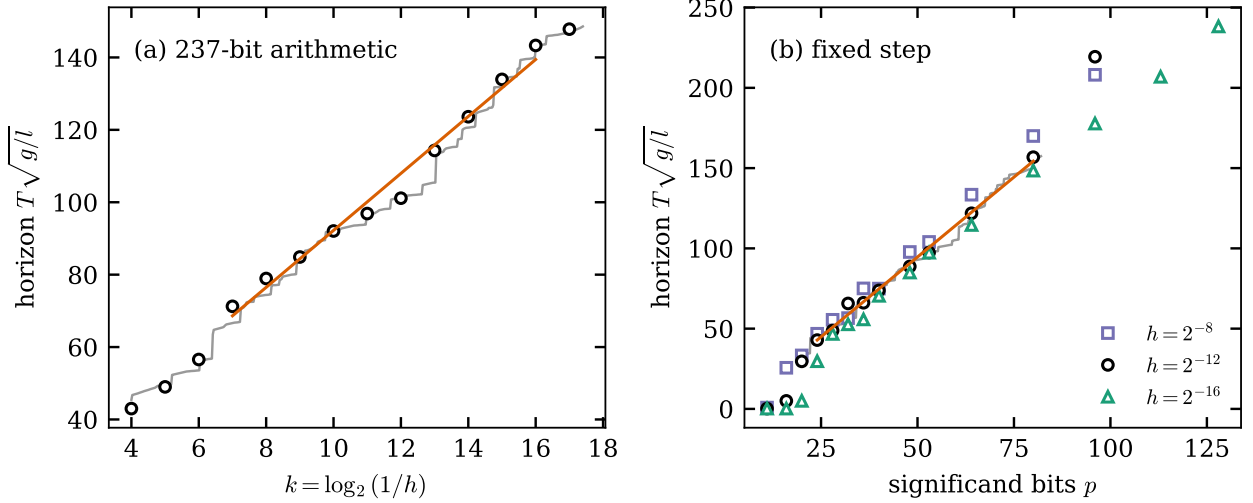


FIG. 3. Horizon T , the first time at which $d \geq 10^{-2}$. (a) Against $k = \log_2(1/h)$ in 237-bit arithmetic, where only truncation error is present. (b) Against the number of significant bits p at fixed step, where only roundoff is present. The straight lines are least-squares fits whose slopes give λ through Eqs. (6) and (7). The gray staircases are the horizons predicted from the measured growth of a perturbation of the initial condition.

total rather than reducing it, and no analysis based on small perturbations describes what happens. This stall is an extreme case of the tendency expressed by the negative sign in Eq. (7).

C. Four measurements of the Lyapunov exponent

Table I collects the estimates. The reference value comes from the two-trajectory method of Benettin *et al.*[21, 22] This method starts a shadow trajectory at a distance $d_0 = 10^{-8}$ from the fiducial trajectory, and once per time unit it accumulates the logarithm of the growth in their separation and pulls the shadow back to d_0 . Averaged over 2×10^4 time units this gives $\lambda = 0.299$ for our initial condition. Of 47 other initial conditions at rest on the same energy surface, 44 are chaotic, with a median of 0.296, and the remaining 3 lie on regular orbits with λ indistinguishable from zero. The direct measurement is the mean slope of the thin line in Fig. 2(b), which is 0.33 over the whole run and 0.35 for $t > 20$, after the slow start from rest.

The two estimates from numerical error agree with each other and with the directly

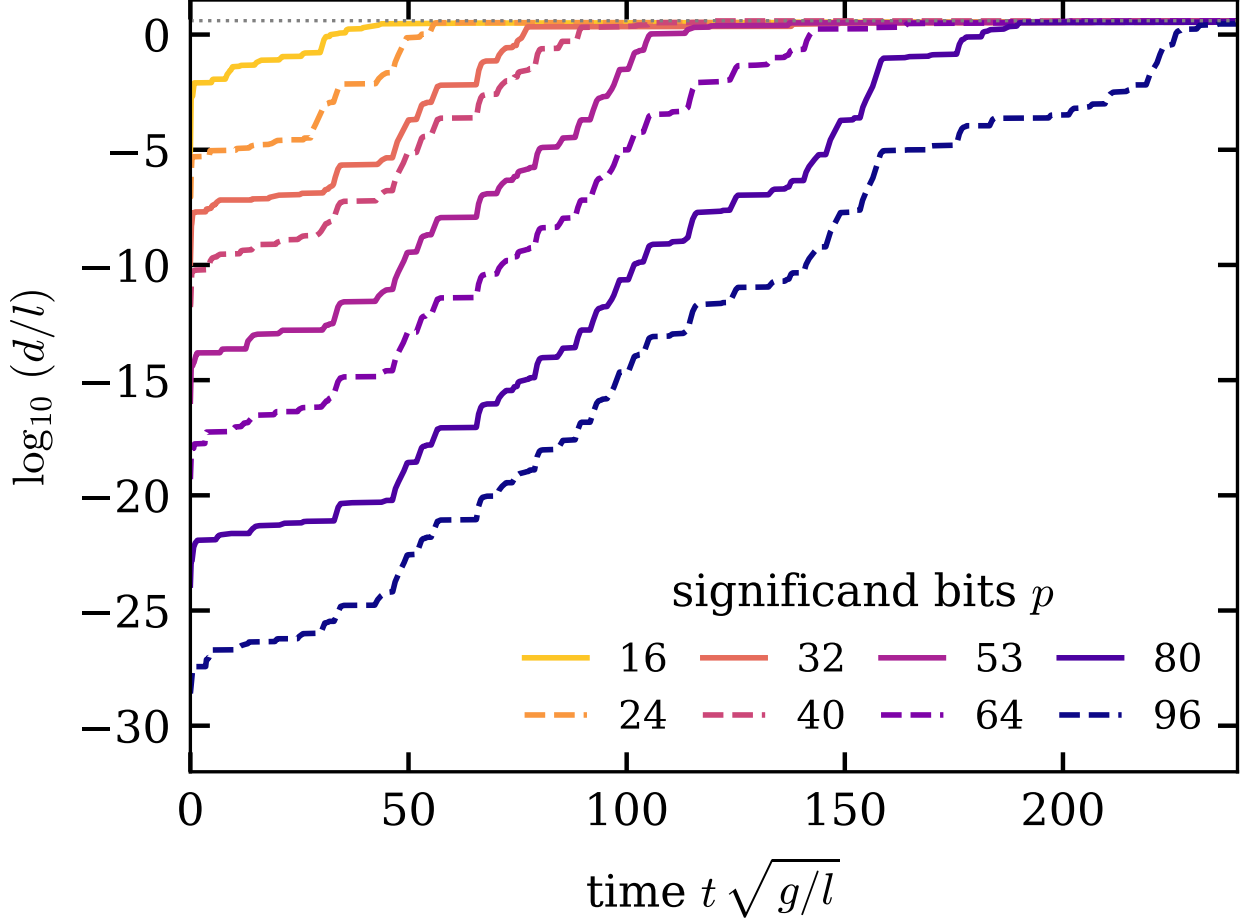


FIG. 4. Separation from the 237-bit run at the same step, $h = 2^{-12}$, for eight significand widths p . Truncation error is common to each pair of runs and cancels, so the curves show the effect of roundoff alone.

TABLE I. The largest Lyapunov exponent of the double pendulum released from rest with both rods horizontal, in units of $\sqrt{g/l}$. The last three rows sample the growth rate over the first 160 time units only, within which it averages 0.28 in the first half and 0.33 in the second. Uncertainties are standard errors of the fitted slopes. At $h = 2^{-8}$ and 2^{-16} the last row becomes 0.31 and 0.34.

Method	λ
Two-trajectory method, 2×10^4 time units	0.299
Growth of a perturbation of 2^{-170}	0.33
Horizon against step, Eq. (6)	0.35 ± 0.02
Horizon against precision, Eq. (7)	0.35 ± 0.01

TABLE II. Longest horizon attainable with RK4 at each precision, and the step at which it occurs. The last column is for rods of length 1 m.

Significand bits p	Best step	$T_{\max}\sqrt{g/l}$	$T_{\max}$ (s)
24 (single)	2^{-7}	53	17
32	2^{-7}	66	21
40	2^{-9}	79	25
53 (double)	2^{-11}	101	32
64 (extended)	2^{-15}	124	40

measured growth of a perturbation over the same interval. All three exceed the long-time value by some 18%. The difference is not an error of method, since a horizon samples the growth rate only up to the horizon. Over its first 160 time units this trajectory happens to be more unstable than on average. The exponent proper is the limit of long times, and the finite-time values scatter about this limit. The distinction is worth making to students, who otherwise picture exponential divergence as uniform. We regard the agreement as the central result of the exercise, because the numerical error that students have been taught to regard as a defect of the computation has measured a property of the physical system.

D. Working precision

Figure 5 shows the third experiment. At large steps every precision gives the same horizon, because truncation dominates and the arithmetic is irrelevant. As the step shrinks each curve leaves the 237-bit line in turn and levels off, and at the lower precisions it then declines, as Eq. (5) predicts. Table II gives the best step and longest horizon for each format.

In double precision the horizon is about 101 time units, or 32 s for a pendulum with one-meter rods, and is reached at $h = 2^{-11}$. A student who responds to a doubtful result by reducing the step to 2^{-17} does 64 times the work, but the horizon is 98, so nothing is gained. In single precision the horizon is 53 time units, reached already at $h = 2^{-7}$, and by $h = 2^{-17}$ it has fallen to 30. Calvão and Penna[5] reported the same turnover for RK4 on this system, where a step of 10^{-4} served best and smaller steps gave less precision.

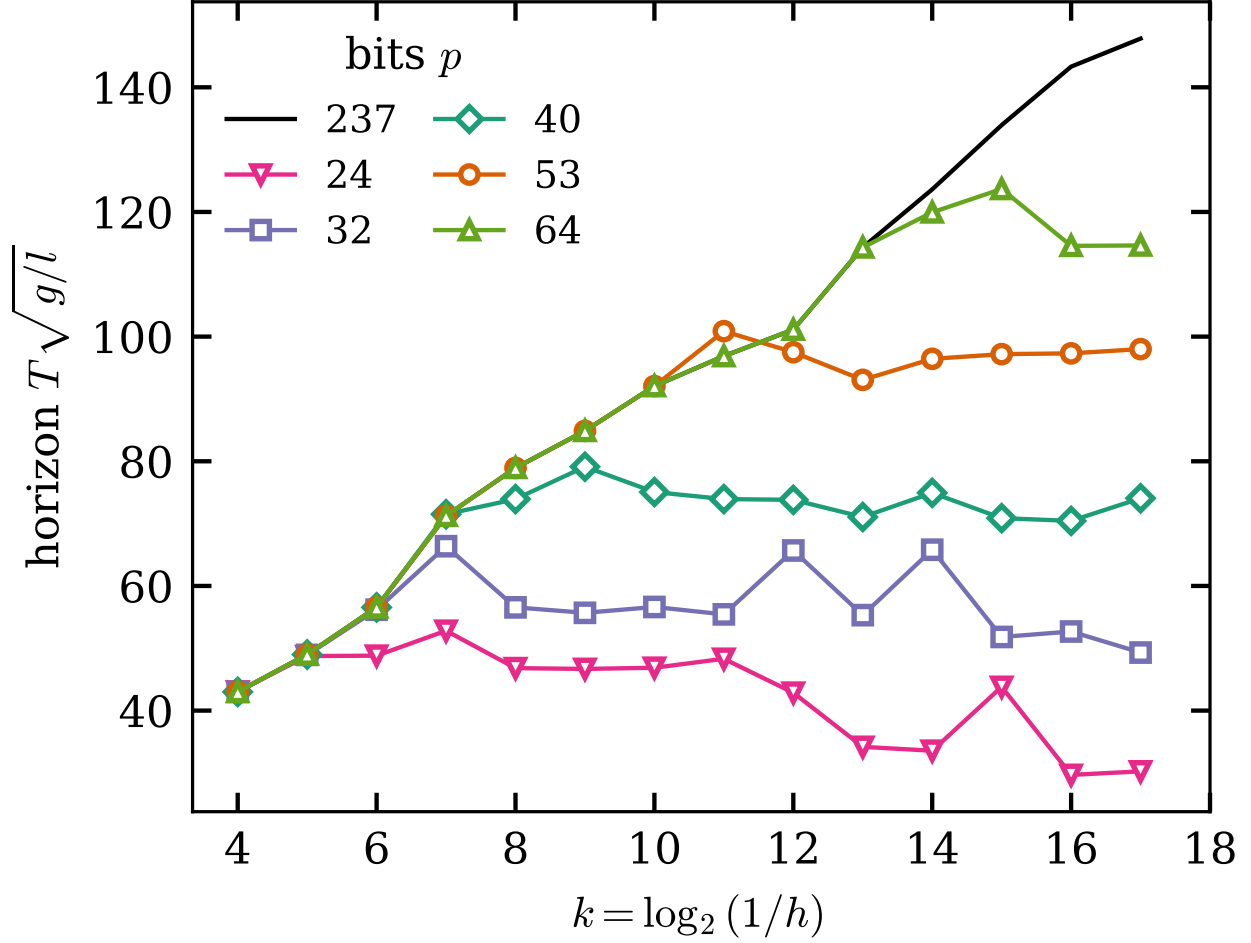


FIG. 5. Horizon against step for five working precisions, with the 237-bit result of Fig. 3(a) for comparison. For each precision there is a step beyond which further refinement does not help.

E. Energy

Figure 6 compares the two errors for an ordinary double-precision run at $h = 2^{-10}$. The error in the energy grows slowly, through the secular drift of a non-symplectic method, and at the end of the run is still below 10^{-10} . Meanwhile, the error in the position reaches the size of the apparatus at $t \approx 92$. The reason is geometric, since energy constrains the state to a three-dimensional surface in the four-dimensional phase space, and the divergence of nearby trajectories takes place within that surface. A computed state can drift an arbitrary distance without leaving this surface.

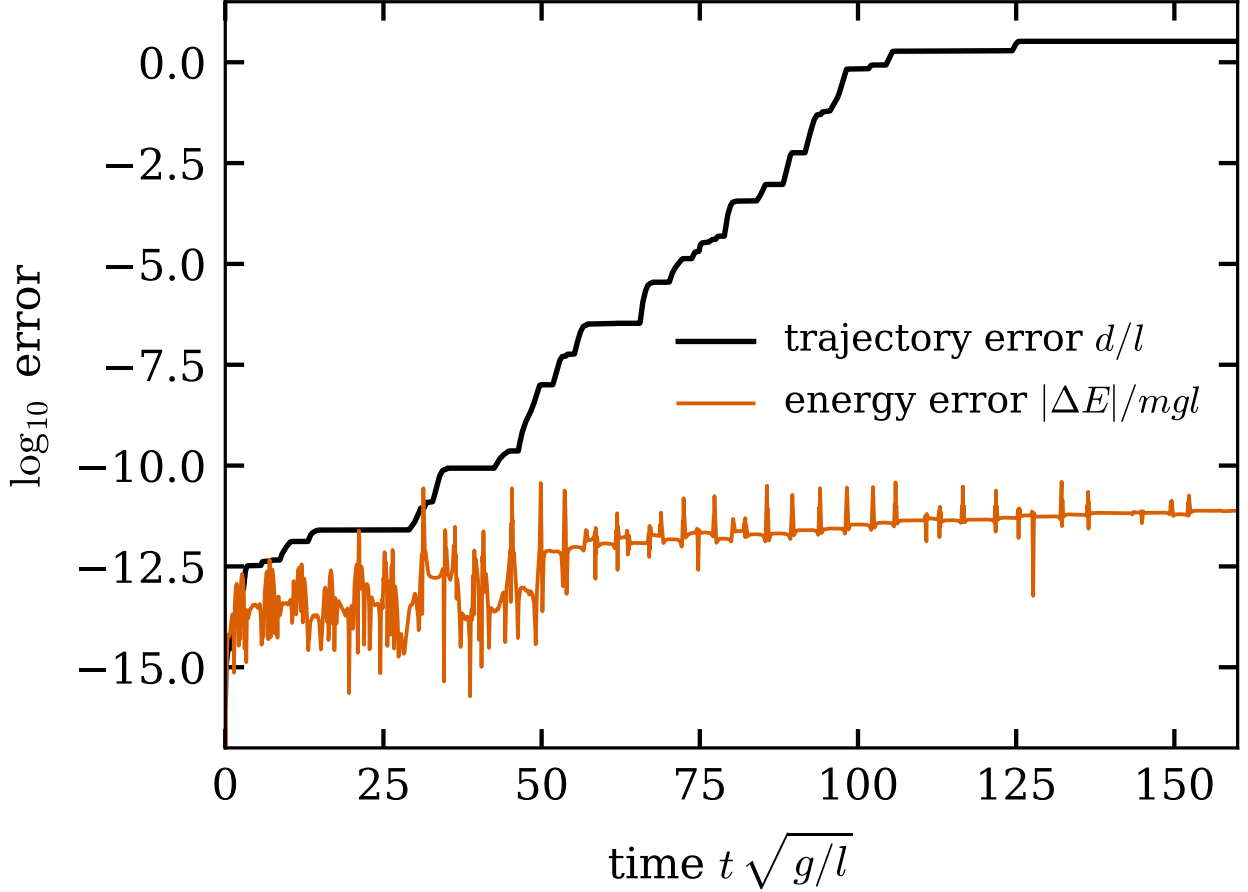


FIG. 6. Error in the position of the lower bob and error in the energy for one run in double precision ($p = 53$) with $h = 2^{-10}$. Energy remains conserved to better than a part in 10^{10} throughout, while the trajectory is lost near $t = 92$.

V. DISCUSSION

A. What accuracy costs

The work of a fixed-step integration is proportional to 2^k . By Eq. (6), extending the horizon by ΔT requires $\Delta k = \lambda \Delta T / (4 \ln 2)$, so the work is multiplied by $e^{\lambda \Delta T / 4}$. To stay on the optimum of Eq. (8), the precision must also rise by about $\lambda \Delta T (4 + \gamma) / (4 \ln 2)$ bits. The appropriate λ depends on the interval, and doubling the double-precision horizon means extrapolating beyond the times sampled, so we use the long-time exponent 0.299. Then, the number of steps is multiplied by roughly $e^{\lambda T_{\max} / 4} \approx 2000$ and about 50 additional bits are needed. At the finite-time rate 0.35 measured over the first 160 time units, the figures would

be nearer 7000 and 59, but either way the horizon grows as the logarithm of the effort. A method of order n replaces the 4 by n , which is the main argument for high-order integrators in this setting. We checked this at low order, where at a fixed early time the error of forward Euler scales as $h^{1.002}$, that of the explicit midpoint rule as $h^{1.999}$, and that of RK4 as $h^{4.000}$. The horizons lengthen in proportion, so forward Euler with $h = 2^{-16}$ is accurate for 31 time units, which is less than the 43 that RK4 achieves with $h = 2^{-4}$ and a thousandth of the work.

In everyday units, the long-time exponent for one-meter rods is $\lambda = 0.94 \text{ s}^{-1}$, so reducing the step tenfold adds $4 \ln 10 / \lambda \approx 10 \text{ s}$ of accurate motion, or 8 s at the finite-time rate, at ten times the cost.

B. What is still right after the horizon

Beyond T , the computed trajectory is not the solution of the initial-value problem posed. However, this trajectory is still meaningful, since shadowing results[15, 23–26] show that, for long times, a noisy computed orbit of a chaotic system stays close to *some* exact orbit with a slightly different initial condition. Hence, properties of the ensemble of orbits rather than of a particular orbit, such as the Lyapunov exponent, are computed correctly by runs hundreds of horizons long. The first row of Table I is such a computation in double precision. We suggest ending a class with the distinction between predicting a trajectory and sampling the dynamics.

C. Use in a course

The exercise suits an upper-division course in computational physics, classical mechanics, or numerical analysis, after students have met RK4 and floating-point numbers. The program is short, since the integrator is the usual RK4 with the number type changed, and all but the two finest reference runs complete in minutes on a laptop. Students with only hardware arithmetic can still do the third experiment, using double precision at the best step of Table II as the reference for single precision. The original version of this project was carried out in a similar way, with decimal steps and a handful of precisions. The logarithmic law was evident even there, since each tenfold reduction of the step added between five and

ten seconds.

VI. SUGGESTED PROBLEMS

1. *Order of the method.* Replace RK4 with forward Euler and the midpoint rule, repeat the first experiment, and show that the slope of T against k is $n \ln 2/\lambda$ for a method of order n . Estimate the step that Euler's method would need to match RK4 at $h = 2^{-10}$.
2. *Energy dependence.* Release the pendulum from rest at smaller angles and show that λ decreases and the horizon lengthens. For sufficiently small angles, show that the error grows as a power of t rather than exponentially, and determine what Eq. (5) becomes.
3. *The roundoff exponent.* Repeat the second experiment at several steps and determine γ . Then, replace the update $\mathbf{y} \leftarrow \mathbf{y} + \Delta\mathbf{y}$ by compensated summation[27] and determine the exponent again.
4. *A symplectic method.* Integrate with the implicit midpoint rule, which is symplectic, and show that the energy error no longer drifts. Compare the horizon with that of the explicit midpoint rule at the same step. Does preserving the geometry of phase space help with predicting the trajectory?
5. *Another system.* Repeat the first and third experiments for the Lorenz equations with the standard parameters, for which $\lambda \approx 0.906$, and predict the double-precision horizon before it is measured.
6. *Shadowing.* Compute the time average of the kinetic energy over 10^4 time units in single and double precision with $h = 2^{-7}$. We find 1.357 and 1.358 in units of mgl , but the trajectories separate after fifty time units. Explain why the two averages agree to a part in a thousand.

SUPPLEMENTARY MATERIAL

The supplementary material contains the Python source for every computation and figure in this paper, the numerical values plotted, and two animations: pendulums differing only in step size, and pendulums differing only in precision, each drawn beside its growing error.

AUTHOR DECLARATIONS

The authors have no conflicts to disclose.

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